

Basic feature

Power	DC5V~DC24V
Temperature MeasuringRange	-30°C~80°C
HumidityMeasuring Range	0~100%RH
Measuring Precision	Temperature:±0.5°C (resolution:0.1°C) Humidity:±5%rh (resolution: 0.1rh)
Output	RS485(ProtocolMODBUSRTU)
Consumption	<0.1W
RS485Communication distance	up to 2000m
Defaultcablelength	1M (can customize length)

SENSORWIRING

Color	Illustrate
Red	VCC
Black	GND
Yellow	485-A
Green ZEElectronic 485-B	

CONFIGURATIONSOFTWAREINSTALLATIONANDUSE

WE PROVIDE A MATCHING "LY485-TOOL", WHICH CAN EASILY USE A COMPUTER TO READ THE PARAMETERS OF THE SENSOR, AND FLEXIBLY MODIFY THE DEVICE ID AND ADDRESS OF THE SENSOR.

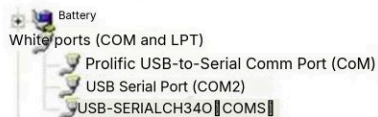
NOTE THAT THERE IS ONLY ONE SENSOR ON THE 485 BUS WHEN USING AUTOMATIC DATA ACQUISITION BY SOFTWARE.

SENSOR CONNECTED TO COMPUTER

The rear sensor is correctly connected to the computer via USB to 485 and provides power.

YOU CAN SEE THE CORRECT CONNECTION IN THE COMPUTER (CHECK THE CONNECTION IN "MY COMPUTER")

PROPERTIES-DEVICES-MANAGER-POR-T")



Open the data package, select "Debugging Software" - "485 Parameter Configuration Software" find the configuration software and open it.

Use of Sensor Monitoring Software

- ①
- The configuration interface is shown in the figure. First, obtain the serial port number and STORE select the correct serial port according to the method in Chapter 3.1.
- ②
- Click "Open serial port" in the software click "Test", you will get the current temperature and humidity, click one key to read, and the basic parameters of the probe will appear.
- ③
- Modify the address and baud rate or calibration according to the needs of use, and at the same time, you can query the current functional status of the device.
- ④
- If the test is unsuccessful, please re-check the equipment wiring and 485 driver installation.

Temperature and Humidity T

Language en-US

Communication Settings

Port Number

Baud Rate 9600

Parity None

Data Bits 8

Stop Bits One

Open Port Close Port Current Datas

Scan Ports Auto Reconnect

Query Address Terminate

485 Address 1

You need to enter the correct 485 address first if you do not know the correct address, please click on the query address

Read/Set Registers

Address

Set

Baud Rate

9600 Set

Humidity Address

Set

Humidity Calibration Temperature Calibration

Set

Set

Read Temperature And Humidity

Temperature Humidity Humidity Address

0 °C 0 rh% 0

Read Stop

Read Cycle 1 sec

☐ Continuous Reading

Data

TemperatureHumidity(RH)Time

Copy Clear

Port State Disconnected Communication State: Abnormal

Date Line: 0

COMMUNICATION BASIC PARAMETERS

Coding	8 bit binary
Data bits	8bit
Parity bit	None
Stop bit	1bit
Error checking	CRC redundant cyclic code
Baud rate	1200bit/s, 2400bit/s, 4800bit/s, 9600bit/s, 14400bit/s, 19200bit/s, Factory default: 9600bit/s (You can modify it yourself through software)

STORE

DATA FRAME FORMAT DEFINITION

Using the Modbus-RTU communication protocol, the format is as follows:

Initial structure≥4 bytes of time

Address code=1byte

Function code=1byte

Data area=Nbytes

Errorcheck=16-bit CRCcode ZElectronic

Time to end structure≥4 bytes

STORE

Address code: the address of the transmitter,which is unique in the communication network (factory defaultx01).

Function code: the function instruction ofthe command sent by the host, this transmitter only uses the function code

0x03(read register data).

Dataarea: The data area is thespecific communicationdata,payattention to the high byte of the 1bits data first!

CRC code: two-byte check code.ZElectronic

Host query frame structure:			STORE		
Address code	Function code	register start address	Register length	Check code low	Check code high
1 byte	1 byte	2 byte	2byte	1 byte	1 byte

Slave acknowledgment frame structure						
address code	Function code	number of valid bytes	data area	second data area	Nth data area	check code
1 byte	1 byte	1 byte	2 byte	2 byte	2 byte	2 byte

REGISTERADDRESS

Register address	PLCor configuration address	Content	Operate	Support function
0000H	40001	Humidity(10times the actual value)	read only	03
0001H	40002	Temperature (10 times the actual value)	read only	03
0100H	40257	Address	read and write	03-06
0101H	40258	Baudrate(1for1200.2 for 2400.3 for 4800.4for 9600.5 for14400.6for19200)	read and write	03-06
0102H	40259	Humidity address	read and write	03-06
0104H	40260	Temperature correction value	read and write	03.06
0105H	40261	Humidity correction value	read and write	03 06

COMMUNICATION PROTOCOL SAMPLES AND EXPLANATIONS

Read the temperature and humidity value of device address 0x01 0x01ic

Example: Read the temperature and humidity value of device address 0x01

Query frame (hexadecimal):

STORE

Address code	Function code	Initial address	Data length	Check code low	Check code high
0x01	0x03	0x00 0x00	0x00 0x02	0xC	0x0B

Response frame (hexadecimal): (For example, the temperature is -9.7°C and the humidity is 48.6%RH)

Address code	Function code	Returns the number of valid bytes	Humidity value	Temperature value	Check code low	Check code high
0x01	0x03	0x04	0x01 0xE6 0xFF 0x9F	0x9F	0x1B	0xA0

Temperature calculation:

When the temperature is lower than 0°C, the temperature data is uploaded in the form of complement code.

Temperature: FF9FH (hex) = -97 = > temperature = -9.7°C

Humidity calculation:

Humidity: 1E6H (Hex) = 486 = > Humidity = 48.6%RH

CHANGE THE DEVICE WITH ADDRESS 01 TO 02

Request frame (hexadecimal):

Address code	Function code	Register address	Contents of temperature calibration	Check code low	Check code high
0x01	0x06	0x01 0x00	0x00 0x02	0x**	0x**

Response frame (hexadecimal):

Address code	Function code	Register address	Contents of temperature calibration value	Check code low	Check code high
0x01	0x06	0x01 0x00	0x00 0x02	0x**	0x**

SET THE BAUD RATE OF DEVICE ADDRESS 01 TO 4800

Change the baud rate of device 01 to 480 (01 means 1200, 01 means 2400, 02 means 4800, 03 means 9600,

04 means 14400, 05 means 19200)

Request frame (hexadecimal):

Address code	Function code	Register address	Baud rate value content	Check code low	Check code high
0x01	0x06	0x01 0x01 0x00 0x01	0x01	0x**	0x**

Response frame (hexadecimal):

Address code	Function code	Register address	Baud rate value content	Low bit of verification code	High bit of verification code
0x01	0x06	0x01 0x01 0x00 0x01	0x01	0x**	0x**

READDEVICEADDRESS

STORE

Query frame (hexadecimal):

Address code	Function code	Initial address	Data length	Check code low	Check code high
0x01	0x03	0x01 0x00	0x00 0x01	0x**	0x**

Response frame (hexadecimal):(for example,the address is read as1)

Address code	Function code	Returns the number of valid bytes	Current address	Check code low	Check code high
0x01	0x03	0x02	0x00 0x01	0x**	0x**

READDEVICEBAUDRATE

Query frame (hexadecimal):

Address code	Function code	Initial address	Data length	Check code low	Check code high
0x01	0x03	0x01 0x01 0x00 0x01 0x*			0x**

Response frame (hexadecimal):For example,the baud rate is 4800 (01means 1200.01 means2400.02means 4800.03 means 9000.04 means 14400, 05 means 19200

Address code	Function code	Returns the number of valid bytes	Current baud rate	Check code low	Check code high
0x01	0x03	0x02	0x00 0x01	0x**	0x**

READ ELECTRONIC HUMIDITY CALIBRATION VALUE

Query frame (hexadecimal):

STORE

Address code	Function code	Initial address	Data length	Check code low	Check code high
0x01	0x03	0x01 0x03	0x00 0x01	0x**	0x**
Address code	Function code	Returns the number of valid bytes	Humidity alibration value	Check code low	Check code high
0x01	0x03	0x02	0x000x01	0x**	0x**

READTEMPERATURECALIBRATIONVALUE

Query frame (hexadecimal):

Address code	Function code	Initial address	Data length	Check code low	Check code high
0x01	0x03	0x01 0x04	0x00 0x01	0x**	0x**
Address code	Function code	Returns the number of valid bytes	temperature calibration value	Check code low	Check code high
0x01	0x03	0x02	0x000x01	0x**	0x**

Common problems and solutions

No output or output error

possible reason:

.The computer has a CoM port, and the selected port is incorrect.

②, the baud rate is wrong.

.The 485 bus is disconnected, or the A and B lines are reversed.

.If the number of devices is too much or the wiring is too long, power supply should be provided nearby, add 485 booster, and increase 120 terminal resistance at the same time.

.The USB to 485 driver is not installed or damaged.

, equipment damage.

Please open the link to download the specific communication protocol and configuration software.

<http://www.liyuanz.com/filedownload/107545>

LY485 Temperature And Humidity Tool

Read/Set Registers

LanguageChoose the correct serial port dress

Communication Settings

Port Number COM4

Baud Rate COM4

Parity None

Data Bits 8

\$t2. Open it

Open Port

Close Port

Scan Ports

Auto Reconnect

Query Address Terminate

485 Address 1

You need to enter the correct 485 address first. If you do not know the correct address, please click on the query address

Baud Rate 9600 4 Set

Humidity Address

Humidity Calibration

Temperature Calibration

Read All

3. Read

Read Stop

Read Cycle 1 sec

Continuous Reading

Read Temperature And Humidity

Temperature Humidity

0 °C 0 rh% 0

Humidity Address

Current Data

Data

TemperatureHumidity(RH% Time)

ZElectronic

STORE

Copy

Clear

Port State: Disconnected Communication State: Abnormal

Date Lines: 0